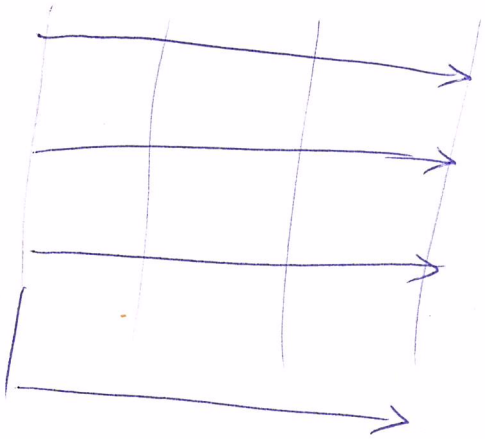
(17) The ray-approximation or geometrical optics

- Problems with wave-based description:
  (- Plane waves do not exist)
  - Complex waves are impossible to compute.
    (Shannon: 2x oversampling in 4D)

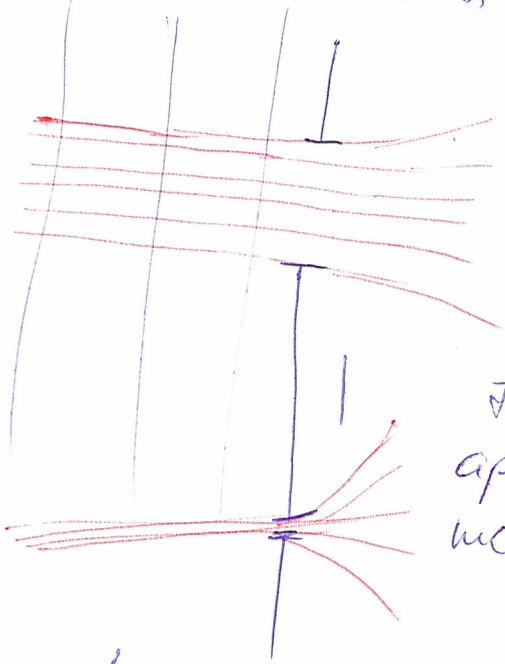
⇓

Ray: The path of energy propagation ($\underline{s}$) in space. ($\underline{s} \parallel \underline{k}$ for isotropic materials)

plane wave ($\underline{s} \perp$ wavefront)

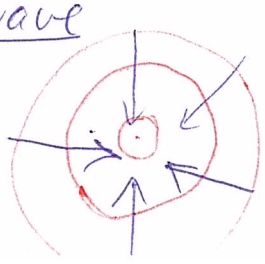


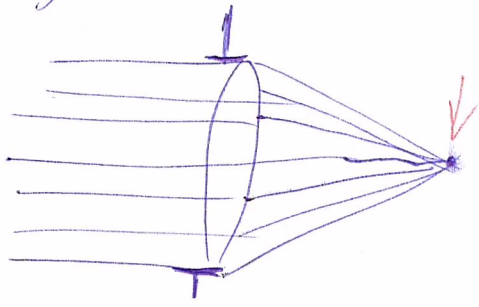
plane wave & opening



Ray approximation can be used for objects $d_{obj} \gg \lambda$, where the effect of diffraction is negligeable. (Zero wavelength)

Spherical wave





Ray tracing is a very straightforward and computationally efficient method.

- propagation in straight line in homogenous media
- reflection and refraction on interfaces
- bending of rays in slowly varying refr. index material (gradient index)

~~Joseph Petzval~~
1807-1891

- Can be efficiently extended to describe wave fronts, abenations and diffraction.

Example:

Spherical mirror: ray tracing of rays with special properties

